Parancsgyűjtemény:

Alapbeállítások:

```
hostname R1
enable secret cisco
banner motd #Authorized Access Only!#
service password-encryption
line console 0
password cisco
login
exit
line vty 0 15
password cisco
login
exit
```

SSH:

```
ip domain-name jedlik.vip
crypto key generate rsa
1024
username admin password class
line vty 0 15
transport input ssh
login local
exit
ip ssh version 2
```

Mac-cím szűrés:

```
int range fa0/0-5
sw m acc
sw port-security
sw por maximum 2
sw por violation restrict/shutdown
sw por mac sticky/address
```

VLAN

Vlanok configurációja:

vlan 10
name guest
exit
vlan 20
name management
exit
vlan 50
name employee
exit
vlan 100
name Native
exit
vlan 120
name Server
exit
vlan 150
name voice
exit
vlan 200
name data
exit

Porthozzárendelés:

int range g5/1, g6/1, g7/1, g8/1
switchport mode access
switchport port-security mac-address sticky
switchport access vlan 10

Voice vlan:

int g9/1
switchport mode access
switchport port-security mac-address sticky
switchport access vlan 20
mls qos trust cos
switchport voice vlan 150

Feketelyuk vlan:

vlan 290
name Black-Hole
exit
int range fa0/18-24

```
sw mode acc
switchport port-security mac-address sticky
switchport access vlan 290
```

Trönkölés:

```
int g0/1
switchport mode trunk
switchport trunk native vlan 100
switchport trunk allowed vlan 10,20,50,100,120,150,200
```

EtherChannel:

```
int range G1/1, G2/1, G3/1, G4/1
channel-group 1 mode on
int port-channel 1
switchport mode trunk
switchport trunk allowed vlan 10,20,50,100,150,200
```

Router-on-a-stick vlan subinterface config:

```
int G6/0
no shut
int G6/0.10
description Default Gateway VLAN 10
encapsulation dot1Q 50
ip add 192.168.10.1 255.255.255.0
ip helper-address 172.16.0.3
exit
int G6/0.20
description Default Gateway VLAN 20
encapsulation dot1Q 20
ip add 192.168.20.1 255.255.255.0
ip helper-address 172.16.0.3
exit
int G6/0.50
description Default Gateway VLAN 50
encapsulation dot1Q 50
ip add 192.168.50.1 255.255.255.0
ip helper-address 172.16.0.3
exit
```

Nativ Vlan Subinterface

```
int G4/0.100
description Nativ Vlan Gateway
encapsulation dot1Q 100 native
ip add 172.16.0.1 255.255.255.0
exit
```

Vlan interface ip:

```
int vlan 100
ip address 192.168.100.3 255.255.255.0
no sh
exit
ip default-gateway 192.168.100.1
```

ROUTING

Statikus routing:

ip route 192.168.10.0 255.255.255.0 10.0.0.2 vagy G0/1

Zero route (az internet felé kimenő port):

```
ip route 0.0.0.0 0.0.0.0 G4/0
```

RIP:

```
router rip
ver 2
no auto
network 172.16.1.0
```

OSPF:

```
router ospf 10
network 192.168.2.0 0.0.0.255 area 0
network 192.168.2.0 0.0.0.255 area 0
auto-cost reference-bandwidth 1000
passive-interface G0/2
redistribute rip
```

FHRP

HSRP config:

Fő router:

```
int
ip add 192.168.1.2 255.255.255.252
standby version 2
standby 1 ip 192.168.1.1
standby 1 priority 120 (100 az alap)
standby 1 preempt
```

Backup router:

```
int
ip add 192.168.1.3 255.255.255.252
standby version 2
standby 1 ip 192.168.1.1
```

Vlan esetén:

Fő router:

```
int G0/0.50
description Default Gateway VLAN 50
encapsulation dot1Q 50
ip add 192.168.50.2 255.255.255.0
ip helper-address 172.16.0.3
standby version 2
standby 1 ip 192.168.1.1
standby 1 priority 120 (100 az alap)
standby 1 preempt
exit
```

Backup router:

```
int G0/0.50
description Default Gateway VLAN 50
encapsulation dot1Q 50
ip add 192.168.50.3 255.255.255.0
ip helper-address 172.16.0.3
standby version 2
standby 1 ip 192.168.1.1
exit
```

GLBP config:

(host-dependent ha csak túlterhelés esetén osztott)
(round-robin ha mindig egyenlő elosztás)

Fő Router:

```
int g0/1
 ip add 192.168.1.2 255.255.255.0
 no sh
exit

glbp 1 timers 5 18
glbp 1 load-balancing host-dependent
glbp 1 priority 120 (100 az alap)
glbp 1 preempt delay minimum 10
glbp 1 name Redundant
glbp 1 ip ip 192.168.1.1
glbp 1 authentication md5 key-string 7 bimbo
```

Backup router:

```
interface g0/1
 ip address 192.168.1.3 255.255.255.0
 no shutdown
exit

glbp 1 timers 5 18
glbp 1 load-balancing host-dependent
glbp 1 preempt delay minimum 10
glbp 1 name Redundant
glbp 1 ip 192.168.1.1
glbp 1 authentication md5 key-string 7 bimbo  ! Same authentication
```

IP NAT

Belső és külső interfész konfigurálása:
(Csak ezek a nat parancsok mennek az interfészekbe a többit a config módba kell írni)

```
interface g0/0/0
 ip address 10.0.0.1 255.255.255.0
 ip nat inside
 no shutdown
```

```
 exit

interface g0/1/0
 ip address 193.226.25.1 255.255.255.0
 ip nat outside
 no shutdown
 exit
```

PAT (a sok belsőt egy külsőre):

Ezt a címlistát natoljuk ki (fordított maszk):

```
access-list 10 permit 10.0.0.0 0.255.255.255
```

Az access listában megadott ipket a g0/1/0 interfész címére fordítja:

```
ip nat inside source list 10 interface g0/1/0 overload
```

Statikus NAT (egy belsőt egy külsőre):

A megadott belső ipt a megadott külsőre fordítja

```
ip nat inside source static 10.0.0.55 193.226.25.12
```

A megadott belső ipről és portról érkező forgalmat a megadott külső címre és portra fordítja

```
ip nat inside source static tcp 10.0.0.55 80 193.226.25.12 80
```

Dinamikus NAT (sok belsőt sok külsőre):

```
access-list 10 permit 10.0.0.0 0.255.255.255
```

A külső címek listája:

```
ip nat pool Kulso 193.226.25.2 193.226.25.30 netmask 255.255.255.224
```

Az access-list 10-et a Kulso poolra fordítja:

```
ip nat inside source list 10 pool Kulso
```

EGYÉB

FTP

```
ip ftp username admin
ip ftp password jelszo
do copy runn ftp
```

DHCP

Dhcp relay:

ip helper-address 172.16.0.3

IPV4 Router DHCP:

conf t
ip dhcp pool 01
network 192.168.0.0 255.255.255.240
default-router 192.168.0.1
exit
ip dhcp excluded-address 192.168.0.1

ip dhcp pool 02
network 192.168.0.16 255.255.255.240
default-router 192.168.0.1
exit
ip dhcp excluded-address 192.168.0.1

Troubleshooting help:

show running-config
show interfaces
show interfaces status
show ip interface brief
show ip route
show vlan brief
show spanning-tree
show mac address-table
show logging